SCJ2013 Data Structure & Algorithms

Introduction to Data Structures & Algorithm

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Objectives:

By the end of the class, students are expected to understand the following:

- problem solving introduction
- algorithm concept
- data structure concept

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Software Eng. & Problem Solving

- Software engineering
  - Provides techniques to facilitate the development of computer program
- Problem solving
  - Taking the statement of a problem and develop a computer program to solve problems.
  - The entire process requires to pass many phases, from understanding the problem, design solution and implement the solution.

Problem Solving

- A solution to a problem is computer program written in programming language which consist of modules.

- Type of Modules:
  - A single, stand-alone function
  - A method of a class
  - A class
  - Several functions or classes working closely together
  - Other blocks of code

Problem Solving

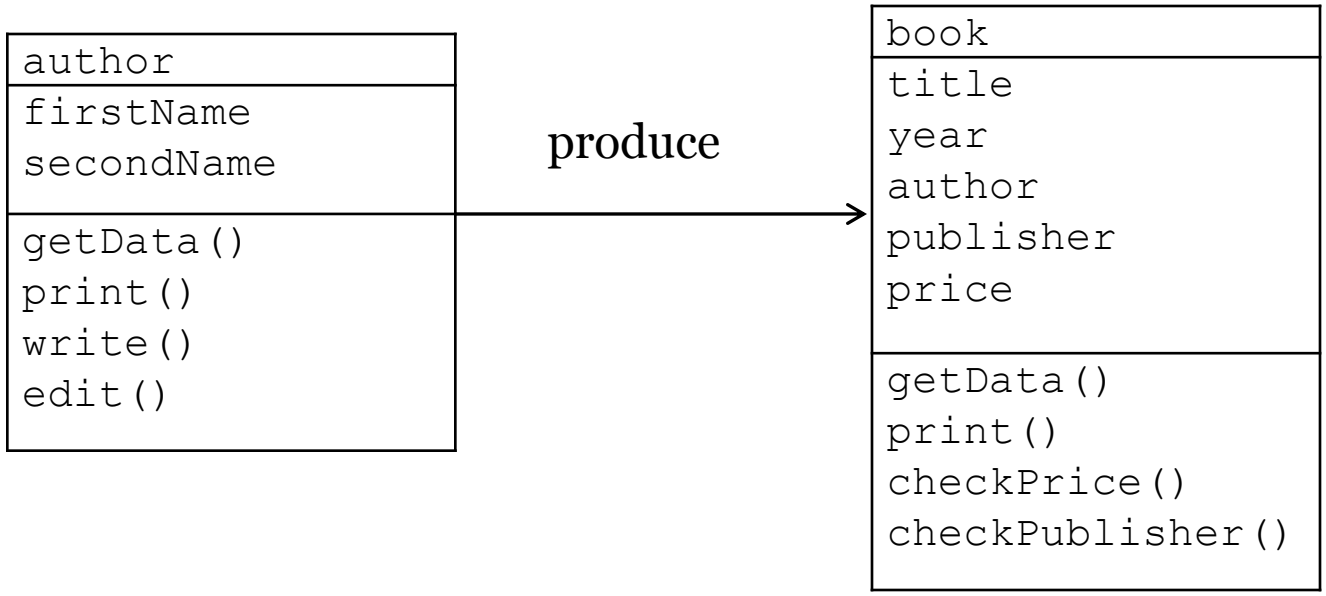
A good solution consists of :

- Modules that
  - organize data collection to facilitate operations
  - must store, move, and alter data
  - use algorithms to communicate with one another

Modularity

- Advantage of module:
  - Constructing programs – small/large modules
  - Debugging programs – task of debugging large programis reduced to small modular program.
  - Reading programs- easier to understand compared to large program
  - Modifying programs – reduce large modification by concentrating on modules
  - Eliminating redundant code – by calling the modules will avoid the same code to be written multiple times

Modularity Example



Algorithm

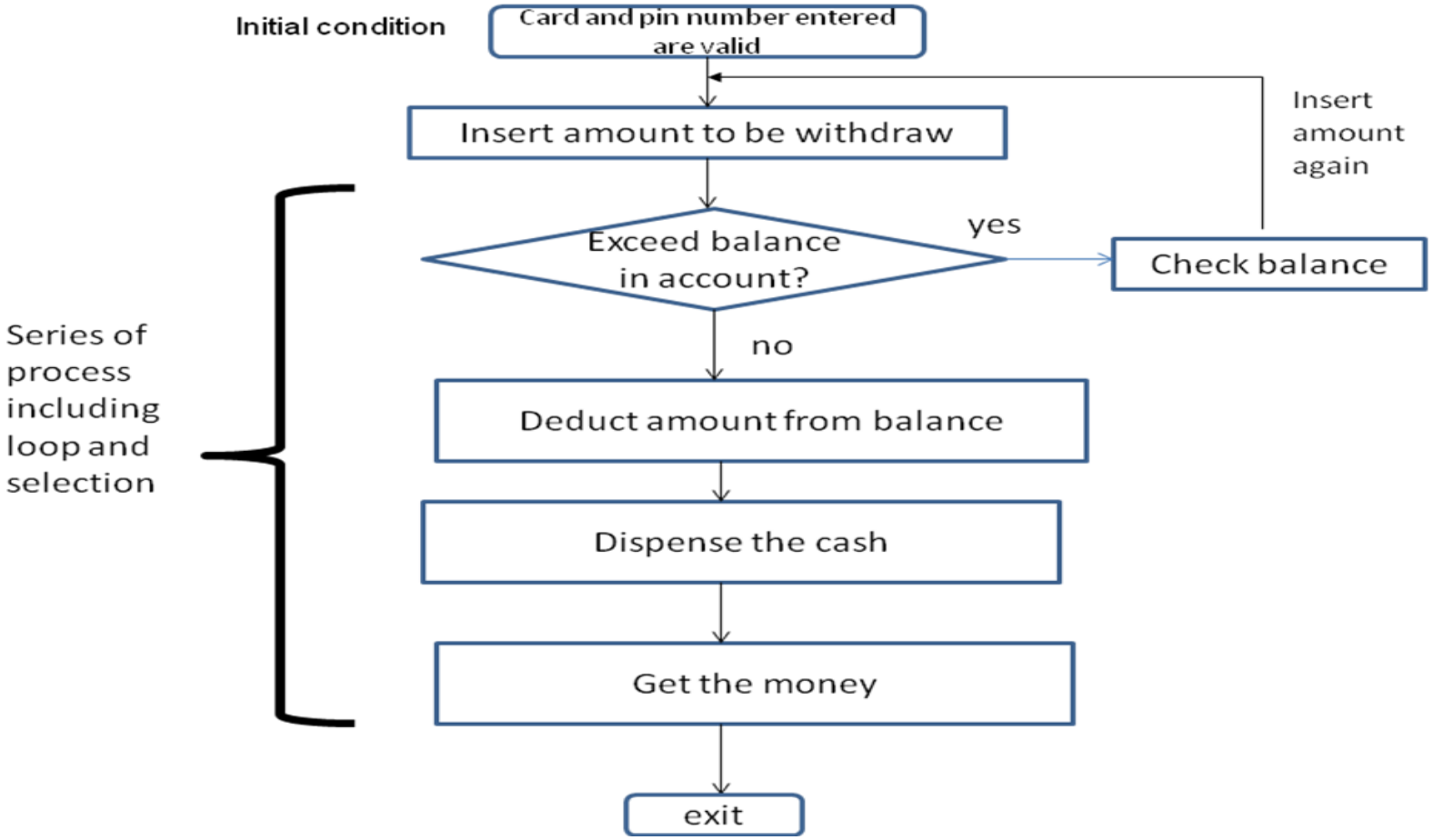
- Module implements algorithms
  - **Algorithm**: a step-by-step recipe for performing a task within a finite period of time
  - **Algorithms** often operate on a collection of data, which is stored in a structured way in the computer memory **(Data Structure)**
  - **Algorithms**: Problem solving using logic

Algorithm

- Well-defined instructions in algorithm includes:

1. when given an initial state, (INPUT)
2. proceed through a well-defined series of successive states, (PROCESS)
3. eventually terminating in an end-state (OUTPUT)

Algorithm


Initial condition
Card and pin number entered are valid
Insert amount to be withdraw
Insert amount again
Exceed balance in account?
yes
Check balance
no
Deduct amount from balance
Dispense the cash
Get the money
exit
Series of process including loop and selection

Algorithm

- **3 types of algorithm basic control structure**
  - Sequential
  - Selection
  - Repeatition (Looping)

Algorithm

- **Basic algorithm characteristics**
  - Finite solution
  - Clear instructions
  - Has input to start the execution
  - Has output as the result of the execution
  - Operate effectively
- Algorithm creation techniques
  - Flowchart, pseudo code, language etc
- Factors for measuring good algorithm
  - Running time
  - Total memory usage

Algorithm & Data Structure

- **Data Structure**
  - A way of storing and organizing data in a computer so that it can be used efficiently
  - Choosing the right data structure will allow the most efficient algorithm to be used
  - A well-designed data structure :
    - allows a variety of critical operations to be performed
    - anable to use few resources, both execution time and memory space, as possible

Data Structure

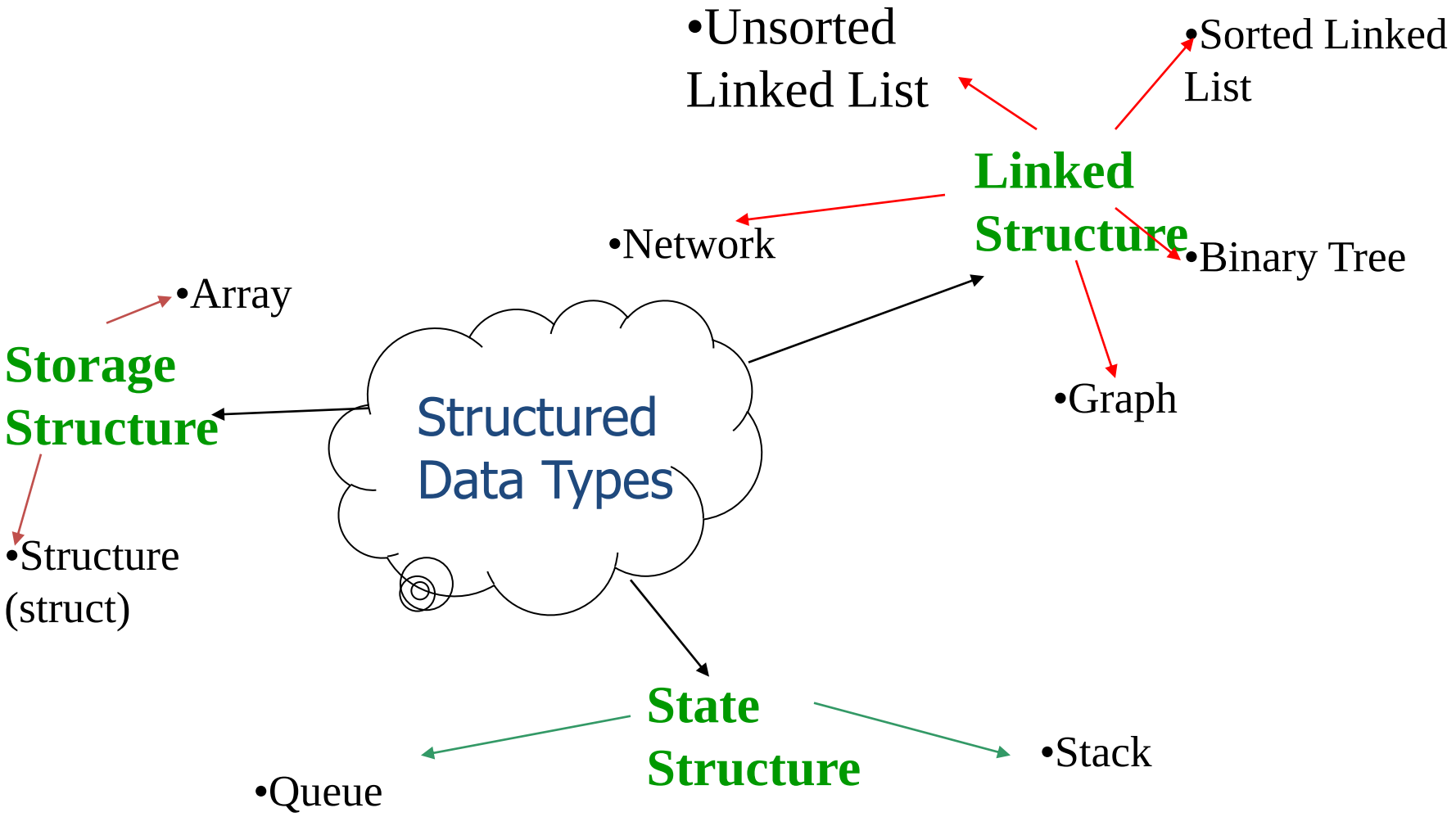
- **Operations to the Data Structure**
  - Traversing- access and process every data in data structure at least once
  - Searching – search for a location of data
  - Insertion – insert item in the list of data
  - Deletion - delete item from a set of data
  - Sorting – sort data in certain order
  - Merging – merge multiple group of data

Data Types

- 2 data types
  1. Basic data types and
  2. structured data types

- Basic Data Types (C++) – store only a single data
  - Integral
    - Boolean – bool
    - Enumeration – enum
    - Character - char
    - Integer – short, int, long
    - Floating point – float, double

Data Types


•Unsorted Linked List
•Sorted Linked List
Linked Structure
•Network
•Binary Tree
•Array
Storage Structure
Structured Data Types
•Graph
•Structure (struct)
State Structure
•Queue
•Stack

Data Types

- Structured Data Types
  - Array – can contain multiple data with the same types
  - Struct – can contain multiple data with different type

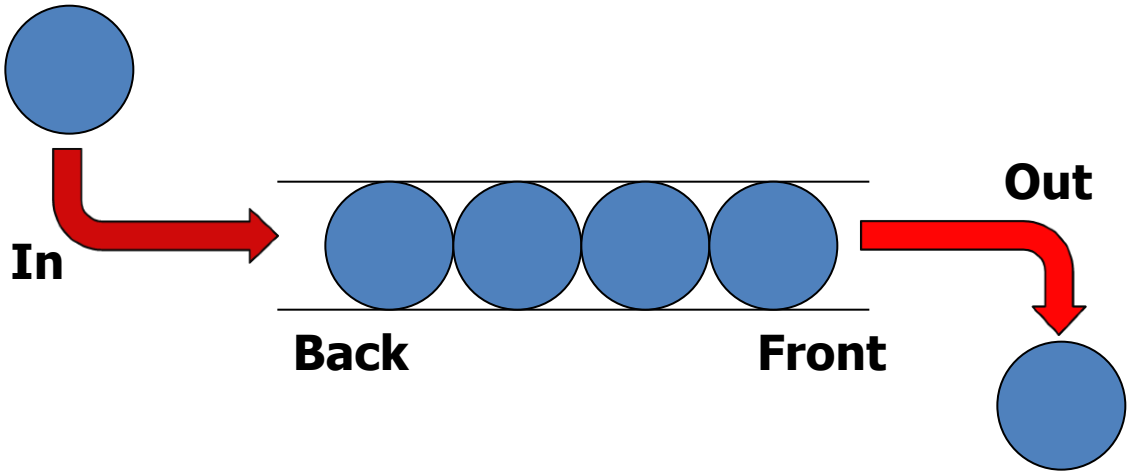
```
typedef struct {
    int age;
    char *name;
    enum {male, female} gender;
} Person;
```

Data Types

- Linked Data Structure
  - Linear Data Structure with restriction
    - Queue & Stack
  - Linear Data Structure with no restriction
    - Unsorted linked list
    - Sorted linked list
  - Non-linear Data Structure
    - Binary Tree
    - Graph

Linear Data Structure with restriction

- Queue
  - First-In-First-Out (FIFO) data structure
  - the first element added to the queue will be the first one to be removed (post office, bank etc)

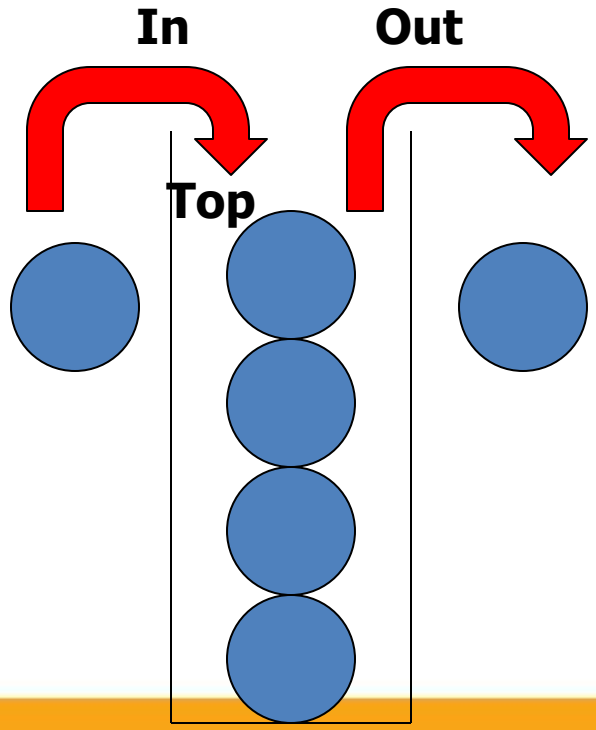


Queue Application



Linear Data Structure with restriction

- Stack
  - Based on the principle of *Last In First Out (LIFO)*
  - Stacks are used extensively at every level of a modern computer system (compiler etc.)

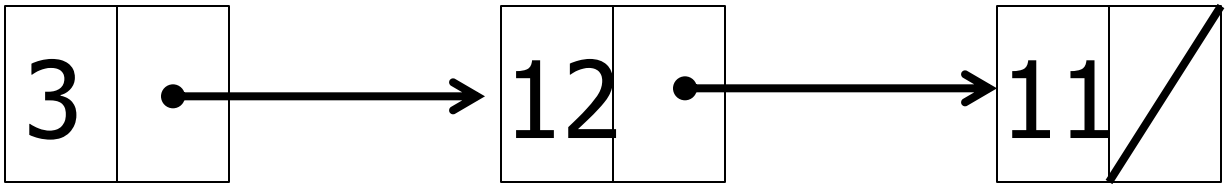


Stack Application



Linear Data Structure with no restriction

- Linked list consists of:
  - a sequence of nodes,
  - data fields
  - one or two links or references pointing to the next and/or previous nodes



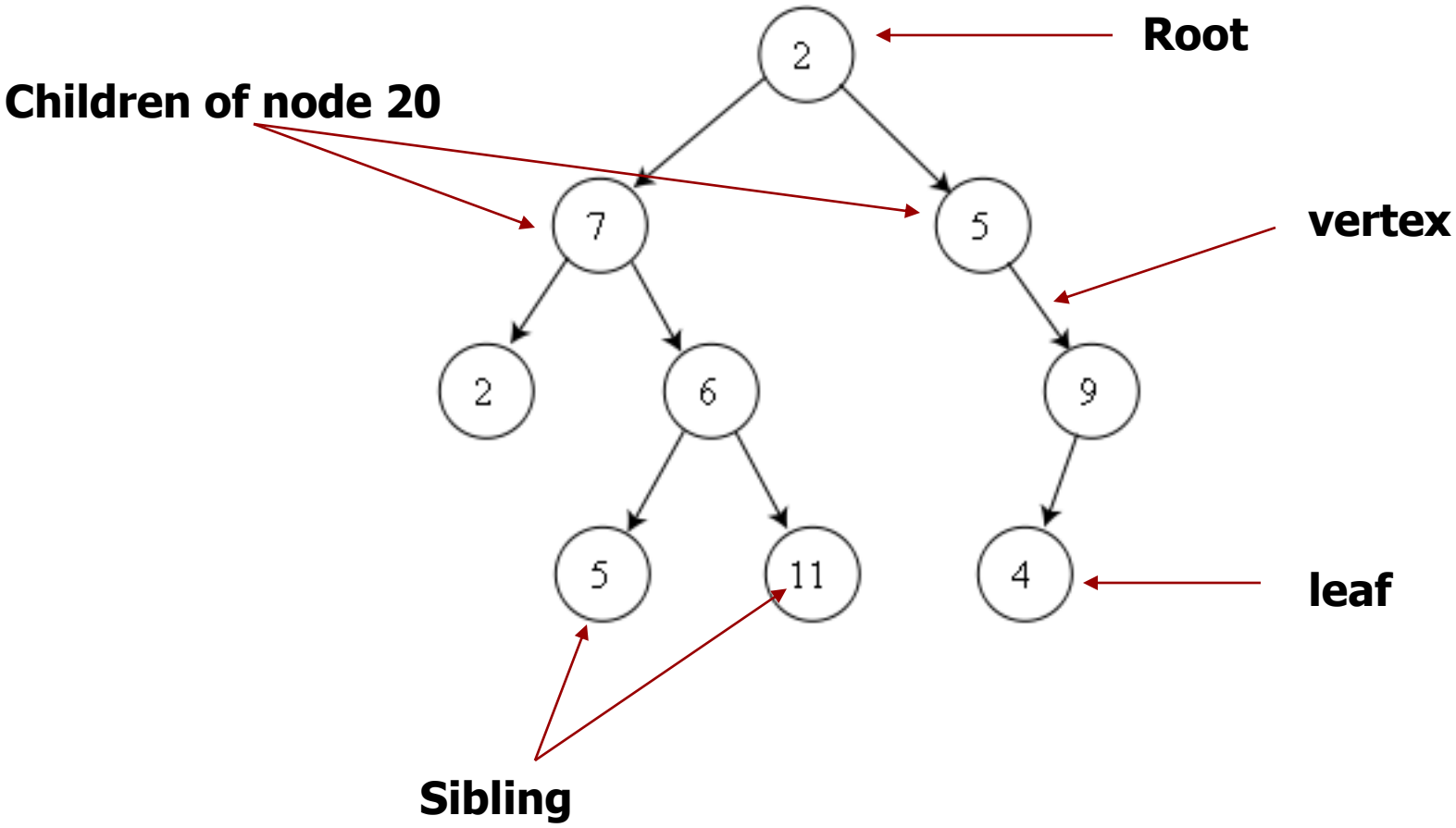
Linear Data Structure with no restriction

- Sorted linked list
  - Data stored in ascending or descending order with no duplicates
  - Insertion at front, middle or rear of the list
  - Deletion will not affect the ascending / descending order of the list
- Unsorted linked list
  - A linked list with no ordering

Non-linear Data Structure

- Tree
  - A data structure based on a tree structure
  - A tree structure is a way of representing the hierarchical nature of a structure in a graphical form
  - a binary tree is a tree data structure in which each node has at most two children
  - Used for searching big amount of data

Tree



Graph

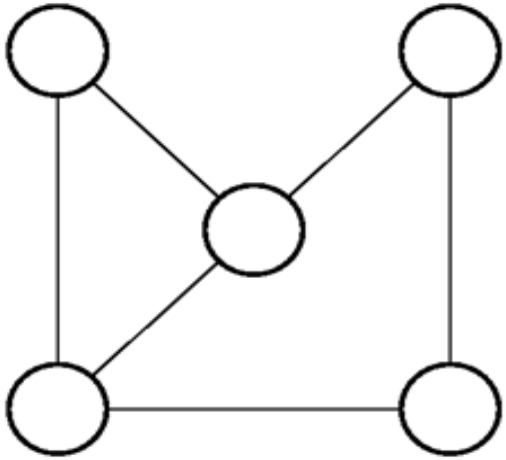
- A graph consists of a set of vertices, and a set of edges, such that each edge is a connection between a pair of vertices.

- Some applications require visiting every vertex in the graph exactly once.

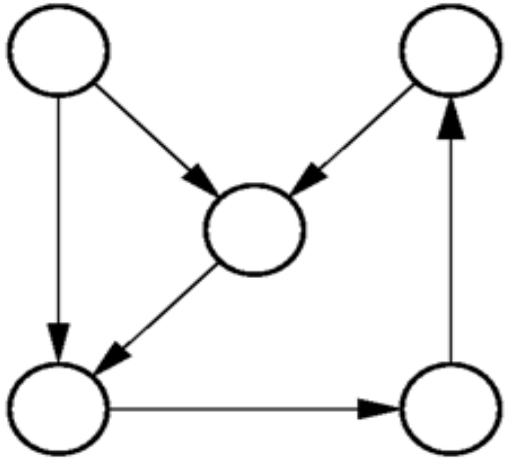
Graph

- The application may require that vertices be visited in some special order based on graph topology.
- Examples:
  - Artificial Intelligence Search (Breadth-first search, depth first search)
  - Shortest paths problems
  - Web sites containing a link to and from other websites.
  - Graph that represent courses and the pre-requisites.

Graph Example



Directed



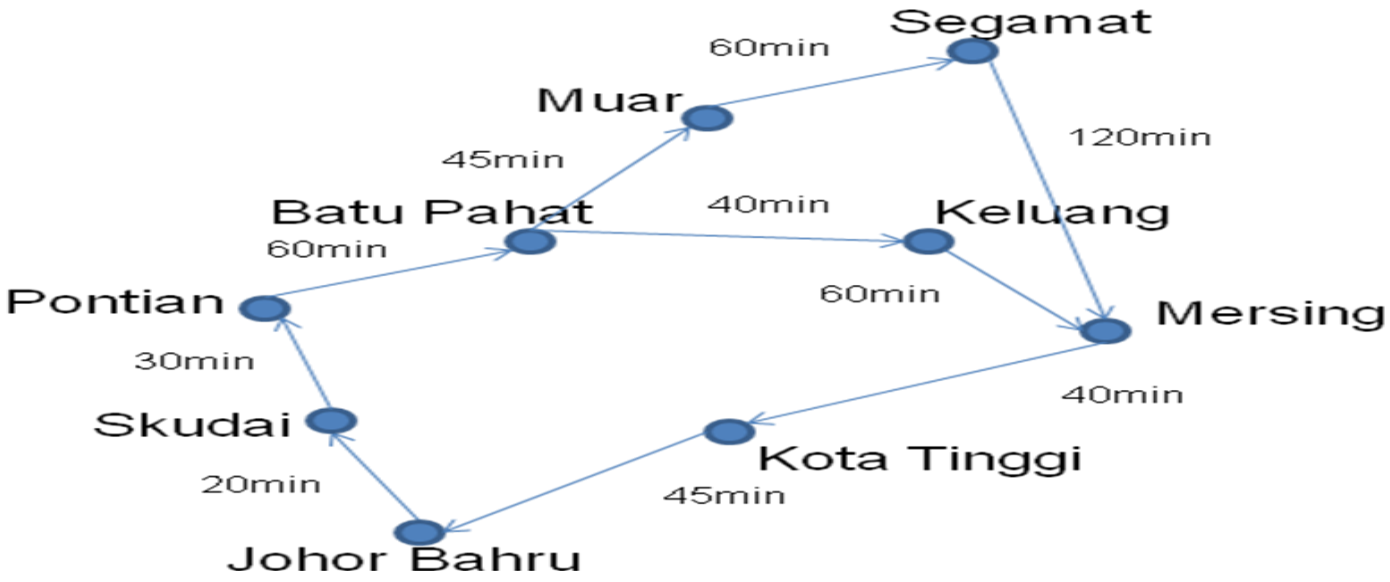
Undirected graph

Network

- Network is a directed graph.
- Can be used to represent a route.
- Example :
  - A route for an airline.
  - A route for delivery vehicles.

Network Example

- Weighted network that represents a route for a delivery truck. The route shows all cities in Johor for the truck to deliver items and the time taken for a journey from one city to another.



Conclusion

In this class you have learned about:

- Problem solving is the entire process of taking the statement of a problem and develop a computer program to solve problems.
- Algorithm is step-by-step recipe for performing a task operate on a collection of data
- Data structure is a way of storing and organizing data in a computer, it allows efficient algorithm to be used
- The knowledge given is to ensure that you are able to provide good solution to problem solving

References

- Frank M. Carano, Janet J Prichard. "*Data Abstraction and problem solving with C++" Walls and Mirrors*. 5th edition (2007). Addision Wesley.
- Nor Bahiah et al. "*Struktur data & algoritma menggunakan C++". Penerbit UTM. 2005.*